

RTL DESIGN OF A 4-BIT MAC UNIT FOR AI ACCELERATOR APPLICATIONS USING VERILOG HDL

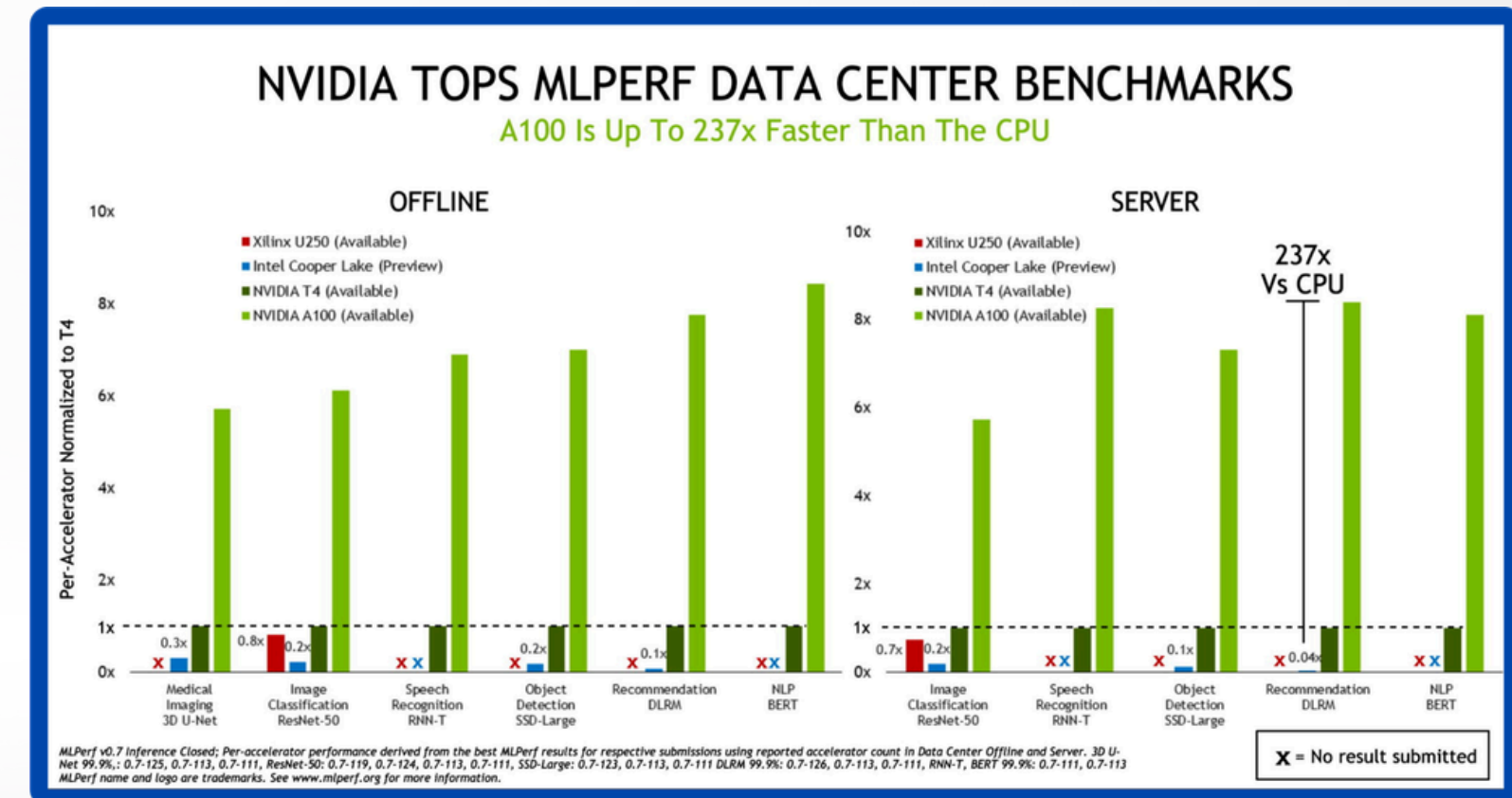
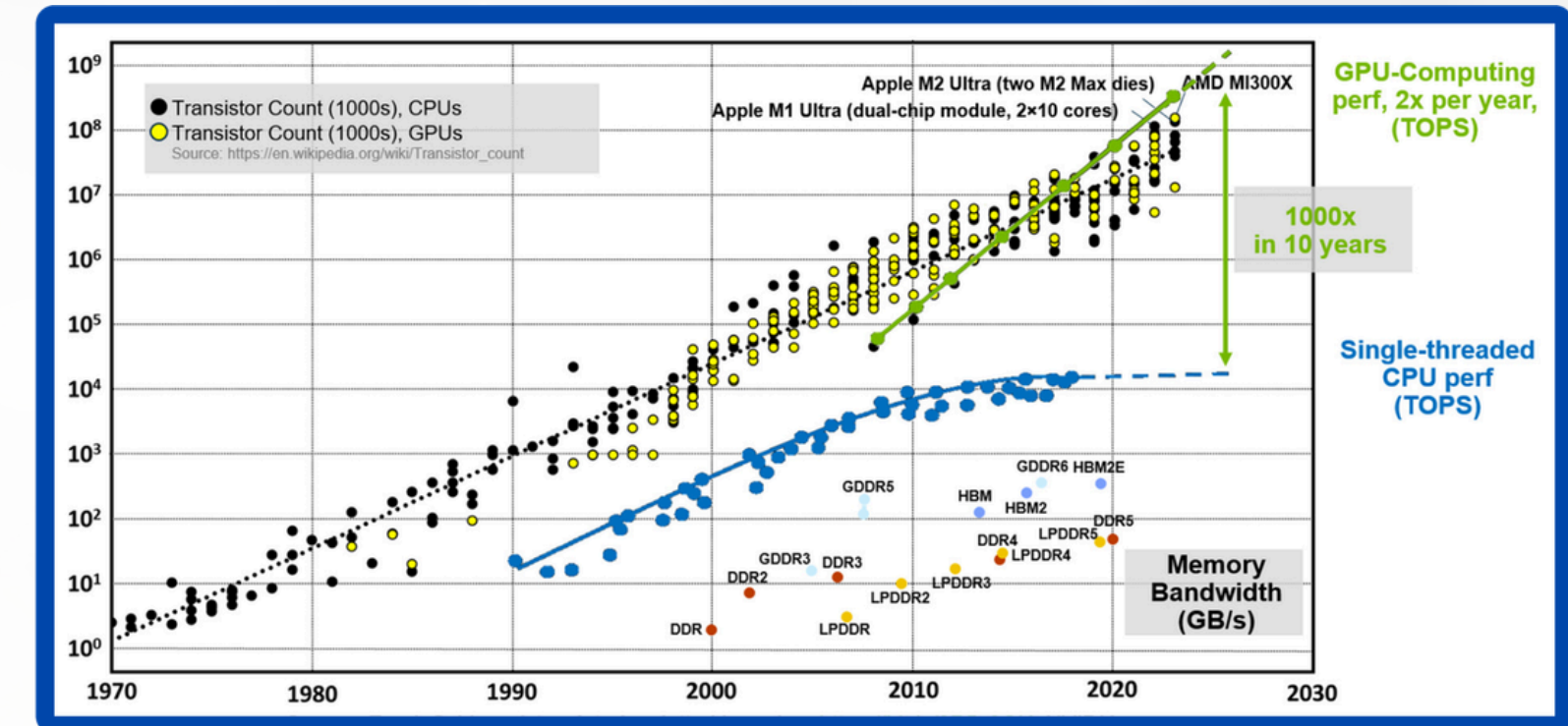
Presented by

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VLSI Design & AI Hardware Research

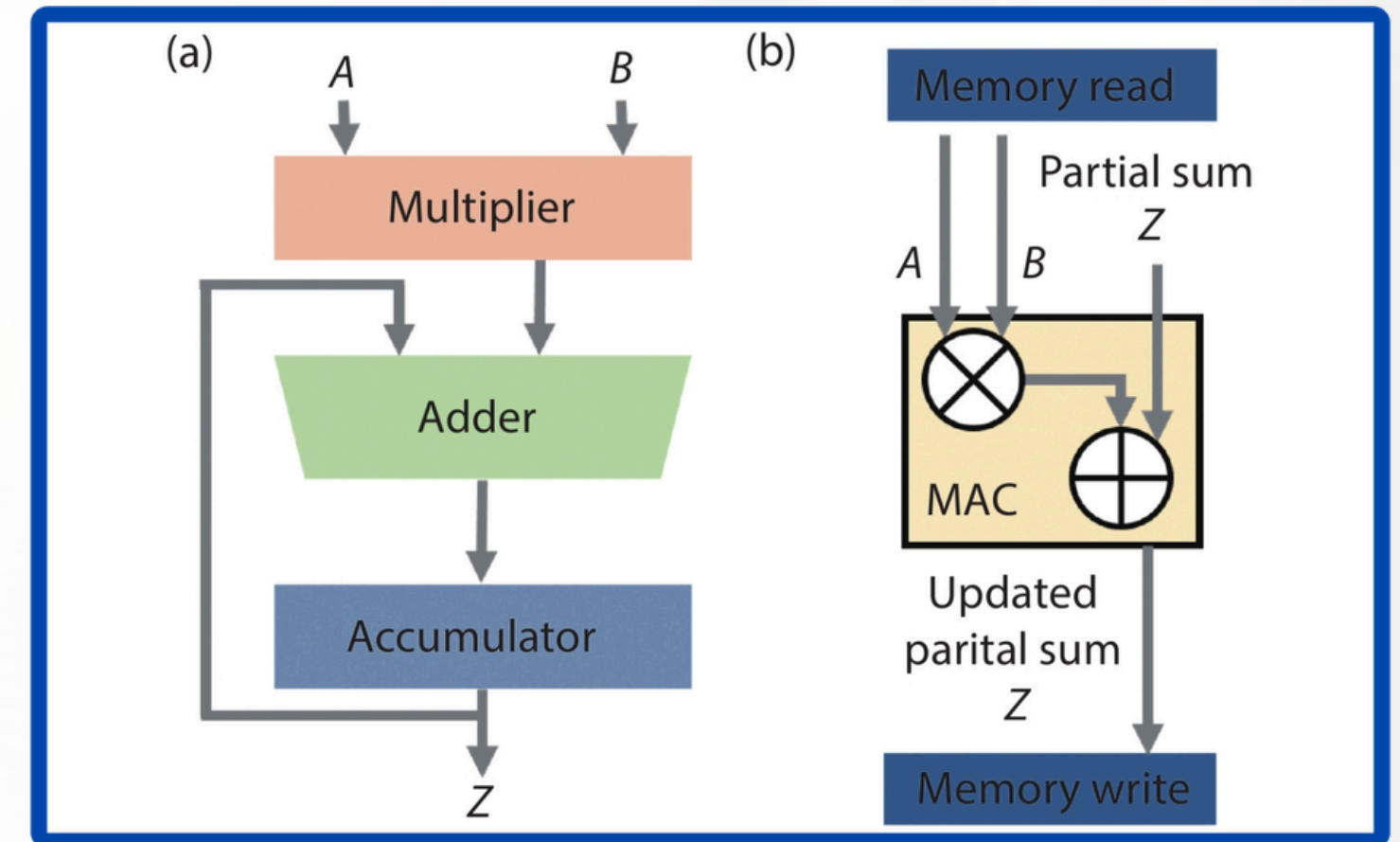
INTRODUCTION TO AI ACCELERATORS

- Artificial Intelligence (AI) applications require massive computations.
- Traditional CPUs are not optimized for parallel AI operations.
- AI Accelerators are specialized hardware architectures designed to improve speed and efficiency for neural network computations.
- These accelerators use parallel processing units such as MAC (Multiply-Accumulate) units.
- AI accelerators are widely used in GPUs, TPUs, Edge AI devices, and deep learning systems.



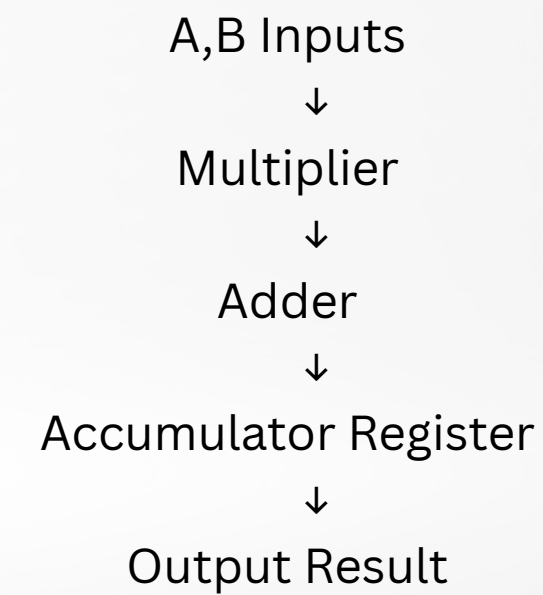
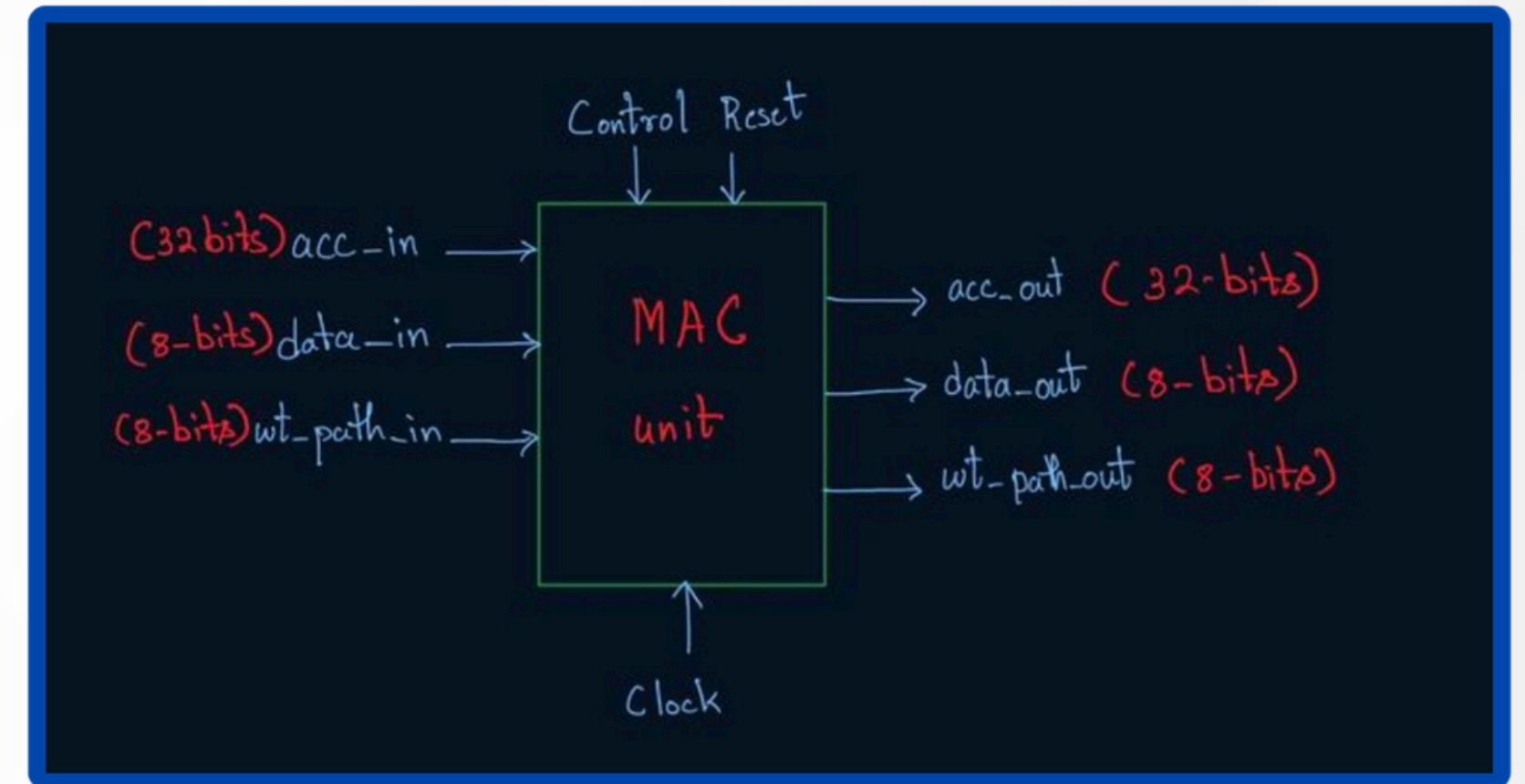
ROLE OF MAC UNIT IN AI ACCELERATORS

- MAC stands for Multiply-Accumulate.
- MAC units perform multiplication and accumulation operations required in neural networks.
- AI accelerators contain multiple MAC units working in parallel.
- MAC operations are heavily used in matrix multiplication and convolution layers.
- The efficiency of MAC units directly affects AI accelerator performance.
- Formula:
$$\text{Result} = \text{Result} + (A * B)$$



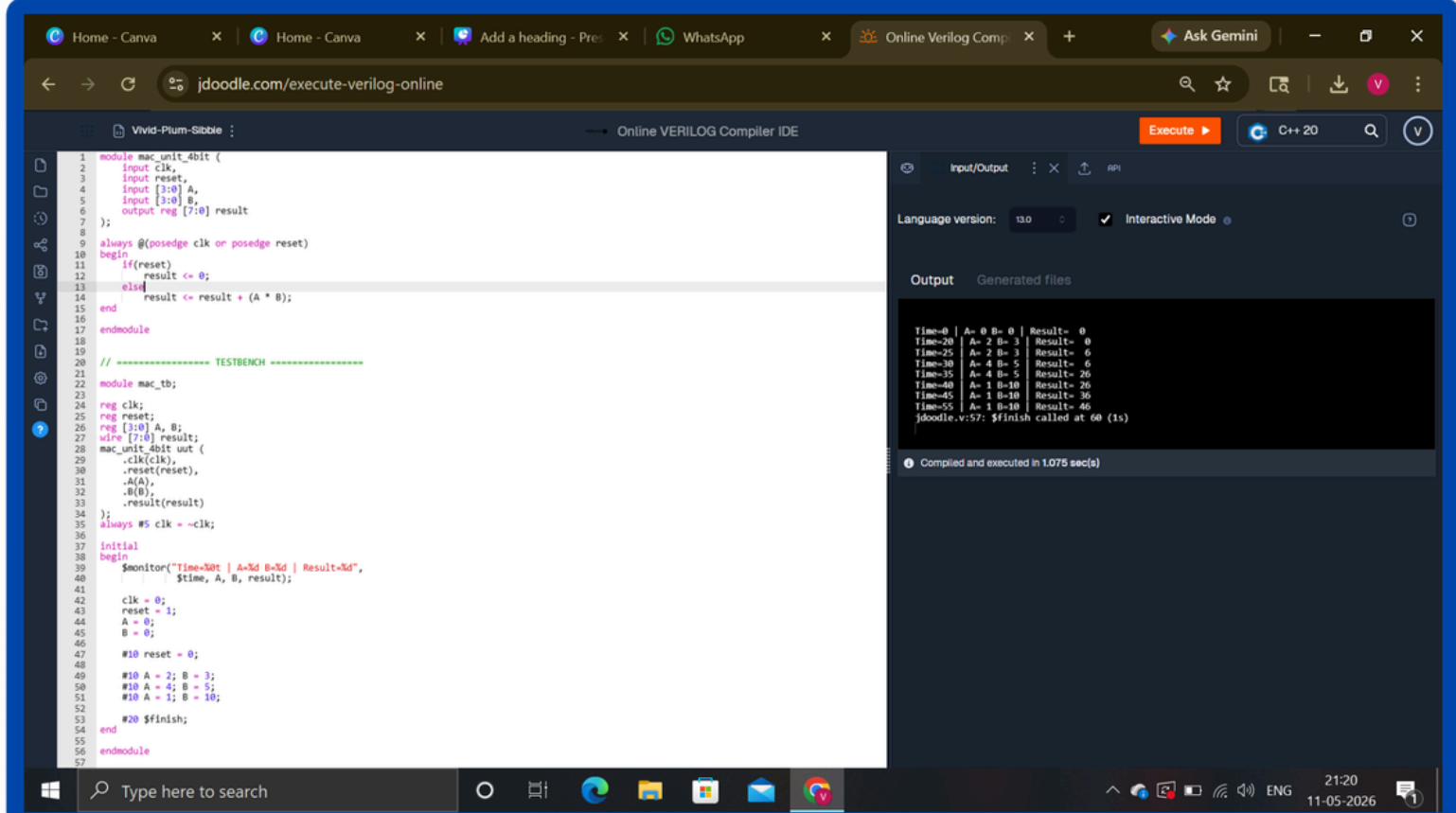
RTL DESIGN OF 4-BIT MAC UNIT

- The MAC unit was designed using Verilog HDL.
- The design consists of:
 - Multiplier
 - Accumulator
 - Register
- Two 4-bit inputs are multiplied and accumulated on every clock cycle.
- A reset signal is used to clear the output register.
- The design operates synchronously with the positive edge of the clock.



VERILOG IMPLEMENTATION OF 4-BIT MAC UNIT

- The MAC unit was implemented using Verilog HDL.
- The design uses synchronous sequential logic.
- Inputs A and B are multiplied and accumulated at every positive clock edge.
- A separate testbench was created to verify functionality.
- Simulation was performed using an online Verilog simulator.
- Different input combinations were applied through the testbench.
- The MAC unit successfully performed multiplication and accumulation operations.
- Output results verified correct MAC functionality.
- Simulation confirmed synchronous operation with clock input.



```
1 module mac_unit_4bit (
2     input clk,
3     input reset,
4     input [3:0] A,
5     input [3:0] B,
6     output reg [7:0] result
7 );
8
9 always @(posedge clk or posedge reset)
10 begin
11     if(reset)
12         result <= 0;
13     else
14         result <= result + (A * B);
15 end
16
17 endmodule
18
19 // ===== TESTBENCH =====
20 module mac_tb;
21
22 reg clk;
23 reg reset;
24 reg [3:0] A;
25 reg [3:0] B;
26 reg [7:0] result;
27
28 mac_unit_4bit uut (
29     .clk(clk),
30     .reset(reset),
31     .A(A),
32     .B(B),
33     .result(result)
34 );
35
36 always #5 clk = ~clk;
37
38 initial
39 begin
40     $monitor("Time=%t | A=%d B=%d | Result=%d",
41             $time, A, B, result);
42
43     clk = 0;
44     reset = 1;
45     A = 0;
46     B = 0;
47
48     #10 reset = 0;
49
50     #10 A = 2; B = 3;
51     #10 A = 4; B = 5;
52     #10 A = 1; B = 10;
53
54     #20 $finish;
55 end
56
57 endmodule
```

Output

Time	A	B	Result
0	0	0	0
10	2	3	6
20	4	5	26
30	1	10	36
40	1	10	46

Compiled and executed in 1.075 secs

APPLICATIONS

AI accelerators are widely used in deep learning and neural network applications.

1

MAC-based architectures are used in GPUs, TPUs, and Edge AI devices.

2

AI accelerators improve computational speed and power efficiency.

3

CONCLUSION

“This project helped me understand the relationship between VLSI design and AI hardware acceleration through practical RTL implementation.”





THANK YOU



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